

$$\omega_n = \frac{1}{\sqrt{LC}} = 2533.5 \text{ rad/s} \quad (1)$$

$$\zeta = \frac{r_L}{2 * \sqrt{\frac{L}{C}}} = 0.0104 \quad (2)$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 2533.3 \text{ rad/s} \quad (3)$$

$$Q_{factor, seriesRLC} = \frac{\omega_n L}{R} = 48.136 \quad (4)$$

$$Q_{3ph} = 3 * V_{ph} * (|I_{ph}|)^2 * X_{ph} = -602.04 \text{ kVAr} \quad (5)$$

$$(6)$$